Portfolio Builder Project

Momentum + Trend Strength

Combined Factor Analysis Report

Factor Testing Framework

Forward Return Horizon: 20 Trading Days

Statistical Correction: HAC / Newey–West

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1 Objective

The objective of this test is to examine whether combining 20-day momentum with 20-day trend strength provides useful cross-sectional information about subsequent stock returns.

The combined signal is constructed using daily cross-sectional standardization of the two component signals. The resulting score is evaluated using:

- Daily decile portfolios
- 20-day future returns
- Information Coefficient (IC)
- Long–short portfolio analysis
- Ordinary statistical tests
- HAC / Newey–West corrected statistical tests

The purpose is to determine whether the combined signal provides evidence of continuation or reversal in subsequent stock returns.

2 Data

The analysis uses the OHLCV stock dataset `Dataset_PS-A.csv`.

Property	Value
Dataset shape	35,616 observations, 15 columns
Number of stocks	24
Valid observations	34,656
Start date	2016-10-05
End date	2022-08-05
Forward return horizon	20 trading days

Table 1: Dataset characteristics used in the combined factor test.

The analysis is performed cross-sectionally for each trading day across the available stocks.

3 Research Hypothesis

The combined factor tests whether recent momentum and trend strength jointly contain information about subsequent stock returns.

The hypothesis is:

> Stocks with different combinations of recent momentum and trend strength characteristics will exhibit different subsequent 20-day returns.

A positive relationship would indicate continuation, where higher combined scores are associated with higher future returns.

A negative relationship would indicate a reversal-type relationship, where higher combined scores are associated with lower future returns.

4 Factor Construction

4.1 20-Day Momentum

Momentum is defined as the return over the previous 20 trading days:

$$MOM_{20,t} = \frac{Close_t}{Close_{t-20}} - 1$$

This measures the recent price performance of each stock.

4.2 Daily Returns

Daily returns are calculated as:

$$r_t = \frac{Close_t}{Close_{t-1}} - 1$$

4.3 20-Day Volatility

The 20-day volatility is the rolling standard deviation of daily returns:

$$VOL_{20,t} = Std(r_{t-19}, \ldots, r_t)$$

4.4 Trend Strength

Trend strength is defined as momentum normalized by recent volatility:

$$TS_{20,t} = \frac{MOM_{20,t}}{VOL_{20,t}}$$

This measures the magnitude of recent price movement relative to the stock's recent volatility.

4.5 Cross-Sectional Standardization

Both momentum and trend strength are standardized cross-sectionally on each date.

For a signal X:

$$Z_{i,t} = \frac{X_{i,t} - \mu_t}{\sigma_t}$$

Thus:

$$Z_{MOM,i,t} = \frac{MOM_{i,t} - \mu_{MOM,t}}{\sigma_{MOM,t}}$$

and

$$Z_{TS,i,t} = \frac{TS_{i,t} - \mu_{TS,t}}{\sigma_{TS,t}}$$

4.6 Combined Signal

The two standardized signals are combined using equal weights:

$$Combined_{i,t} = Z_{MOM,i,t} + Z_{TS,i,t}$$

A higher combined score therefore represents a stock with a relatively high combination of momentum and trend strength.

5 Future Return Construction

The dependent variable is the subsequent 20-trading-day return:

$$R_{i,t,t+20} = \frac{Close_{i,t+20}}{Close_{i,t}} - 1$$

The same 20-day forward-return horizon is used throughout the factor research framework to maintain comparability.

6 Signal Distribution

The combined signal has the following distribution:

Statistic	Combined Score
Count	34,656
Mean	0.000000
Std. Dev.	1.926599
Minimum	-6.460668
25th Percentile	-1.369688
Median	-0.087198
75th Percentile	1.316333
Maximum	7.367229

Table 2: Distribution of the combined momentum-trend-strength signal.

The mean is approximately zero because the component signals are standardized cross-sectionally.

7 Decile Analysis

Stocks are sorted into ten cross-sectional deciles each day according to the combined signal.

G1 contains stocks with the lowest combined scores, while G10 contains stocks with the highest combined scores.

Decile	Mean Future 20-Day Return
G1	2.1166%
G2	2.2849%
G3	1.9100%
G4	2.0956%
G5	1.7807%
G6	1.6871%
G7	1.6640%
G8	1.1991%
G9	1.3987%
G10	0.7250%

Table 3: Average future 20-day return by combined-signal decile.

The lowest-score group, G1, has a future 20-day return of 2.1166%, while the highest-score group, G10, has a future 20-day return of only 0.7250%.

The direct G1–G10 spread is therefore:

$$2.1166\% - 0.7250\% = 1.3916\%$$

The long–short portfolio calculation gives the same spread:

$$\overline{LS} = 1.3916\%$$

The decile pattern is predominantly downward, although it is not perfectly monotonic. In particular, G2 and G4 are higher than G1, and G9 is higher than G8.

Nevertheless, the highest-score group G10 has substantially lower future returns than the lowest-score group G1.

7.1 Decile Return Graph

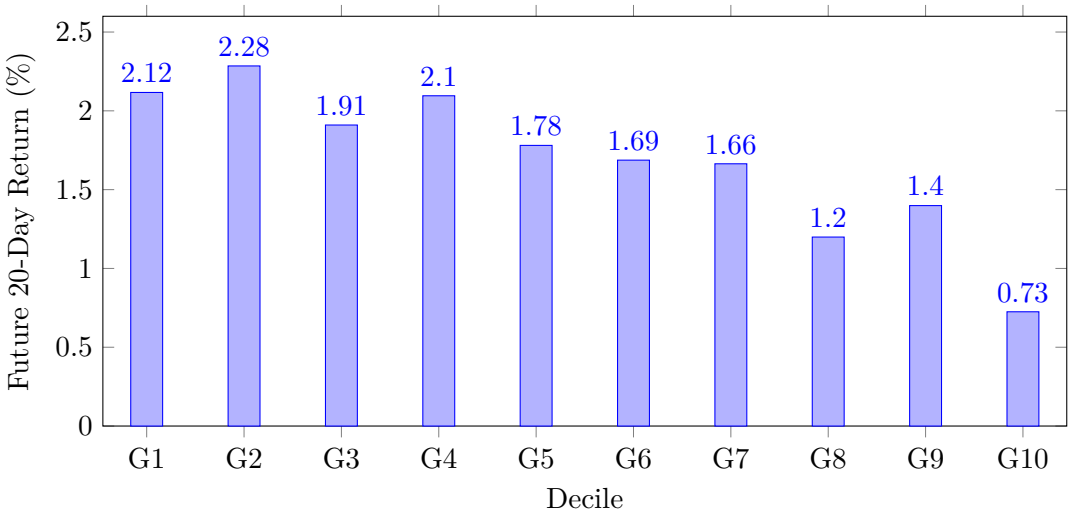


Figure 1: Future 20-day returns across Momentum + Trend Strength deciles.

8 Information Coefficient (IC) Analysis

The Information Coefficient measures the daily cross-sectional Spearman rank correlation between the combined signal and future 20-day returns.

The results are:

Metric	Value
Mean IC	-0.068463
Median IC	-0.072174
IC Std. Dev.	0.245831
Positive IC Fraction	38.85%
IC t-statistic	-10.582904
IC p-value	< 0.001

Table 4: Information Coefficient statistics.

The mean IC is:

$$\overline{IC} = -0.068463$$

The negative sign indicates that higher combined scores tend to be associated with lower subsequent 20-day returns.

The positive IC fraction is only 38.85%, indicating that negative daily cross-sectional relationships occur more frequently than positive ones.

The ordinary IC test gives:

$$t = -10.582904$$

with:

$$p < 0.001$$

This provides strong evidence of a negative cross-sectional relationship under the ordinary IC test.

9 Long–Short Portfolio Analysis

The long–short portfolio is constructed as:

$$LS_t = R_{G1,t} - R_{G10,t}$$

The portfolio therefore buys the lowest combined-score decile and shorts the highest combined-score decile.

The results are:

Statistic	Value
Mean daily return	1.3916%
Median daily return	1.0491%
Daily standard deviation	6.7859%
Annualized return	350.6955%
Annualized volatility	107.7230%
Sharpe-like ratio	3.2555
Ordinary t-statistic	7.793008
Ordinary p-value	< 0.001

Table 5: Long–short portfolio statistics.

The mean long–short return is:

$$\overline{LS} = 0.013916$$

or:

$$1.3916\%$$

per daily portfolio observation.

The ordinary t-test gives:

$$t = 7.793008$$

with:

$$p < 0.001$$

indicating strong statistical significance under the ordinary independence assumption.

9.1 Long–Short Statistics Graph

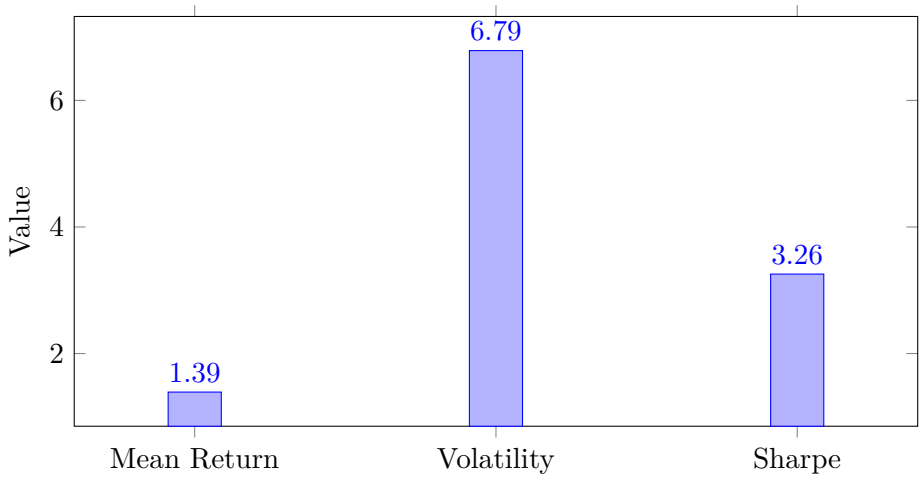


Figure 2: Selected long–short portfolio statistics.

10 Overlapping Return Problem

The analysis uses a 20-trading-day forward return.

Therefore, consecutive observations have overlapping return windows.

For example:

$$R_{t,t+20}$$

and

$$R_{t+1,t+21}$$

share most of their return period.

This creates serial dependence in the sequence of long–short portfolio observations.

As a result, ordinary t-tests can underestimate the true standard error and overstate statistical significance.

Therefore, a HAC / Newey–West correction is required for the main inference.

11 HAC / Newey–West Test

A HAC / Newey–West standard error with 20 lags is used to account for serial dependence caused by the overlapping 20-day forward returns.

The results are:

Statistic	Value
Mean return	0.013916
HAC standard error	0.004991
HAC t-statistic	2.788099
HAC p-value	0.005302

Table 6: HAC / Newey–West test results.

The HAC-adjusted t-statistic is:

$$t_{HAC} = 2.788099$$

and:

$$p_{HAC} = 0.005302$$

Since the p-value is below 0.05, the long–short relationship remains statistically significant after accounting for the overlapping-return problem.

This is the key statistical result of the test.

12 Comparison of Statistical Tests

Test	t-statistic	p-value
IC test	-10.582904	< 0.001
Ordinary Long–Short	7.793008	< 0.001
HAC / Newey–West	2.788099	0.005302

Table 7: Comparison of statistical tests.

The ordinary tests indicate strong statistical significance. After accounting for serial dependence, the t-statistic decreases from 7.793008 to 2.788099.

However, the HAC p-value remains below 0.05.

Therefore, the statistical significance is not solely a result of the overlapping-return structure.

13 Economic Interpretation

The Momentum + Trend Strength signal exhibits a negative relationship with subsequent 20-day stock returns.

The lowest combined-score group produces substantially higher future returns than the highest combined-score group.

This is consistent with a reversal-type relationship rather than a conventional momentum continuation relationship.

The mean IC of:

$$-0.068463$$

provides evidence of a negative cross-sectional relationship.

The long–short portfolio produces:

$$1.3916\%$$

mean return per daily portfolio observation.

Most importantly, the HAC test remains statistically significant:

$$t_{HAC} = 2.788099$$

$$p_{HAC} = 0.005302$$

Thus, the observed relationship survives the correction for overlapping forward returns.

The decile relationship is not perfectly monotonic, but the extreme deciles show a substantial difference, with G1 at 2.1166% and G10 at 0.7250%.

14 Comparison with Previous Combined Signals

The Momentum + Trend Strength combination can be compared with the two previously tested combined signals.

Metric	Mom + Rel. Volume	Mom + Volatility	Mom + Trend Strength
Mean IC	-0.056431	-0.058743	-0.068463
Long–Short Return	1.0499%	0.9434%	1.3916%
Sharpe-like Ratio	2.5731	2.4496	3.2555
HAC t-statistic	2.7743	2.2361	2.7881
HAC p-value	0.005531	0.025345	0.005302

Table 8: Comparison of the three Stage 3 combined signals.

The Momentum + Trend Strength combination has the most negative mean IC among the three combinations:

$$-0.068463$$

It also has the largest long–short mean return:

$$1.3916\%$$

and the largest Sharpe-like ratio:

$$3.2555$$

Its HAC t-statistic is:

$$2.7881$$

with:

$$p = 0.005302$$

Thus, the combination shows strong evidence of a reversal-type relationship in the current sample.

However, this comparison does not establish that Trend Strength adds independent information to Momentum. An incremental or nested-factor analysis would be required to establish additional predictive information.

15 Interpretation of Annualized Return

The reported annualized return is:

$$350.6955\%$$

This is obtained mechanically by multiplying the mean daily long–short return by 252:

$$0.013916 \times 252 = 3.506955$$

or:

$$350.6955\%$$

This figure should **not** be interpreted as an expected annual investment return.

The underlying dependent variable is a 20-day forward return, which creates substantial overlap between consecutive observations.

Therefore, the annualized return is included only as a mechanical descriptive statistic for consistency with the research framework.

16 Limitations

The following limitations should be considered:

1. The forward return horizon is 20 trading days, causing substantial overlap between consecutive observations.
2. The ordinary t-test therefore assumes more independent information than is actually present.
3. The HAC / Newey–West test is therefore the more appropriate inference for the long–short portfolio.
4. The decile pattern is not perfectly monotonic.
5. The combined signal uses equal weights for momentum and trend strength. Other weighting schemes were not tested.
6. Transaction costs, market impact, short-selling costs, and other implementation costs are not included.
7. The annualized return is a mechanical transformation and should not be interpreted as a realistic annual expected return.
8. The combined test alone does not establish whether trend strength provides incremental predictive information beyond momentum.

17 Factor Classification

Based on the results, the Momentum + Trend Strength combination can be classified as a:

Significant Combined Reversal Signal

The main evidence supporting this classification is:

- Mean IC = -0.068463
- Median IC = -0.072174
- G1 future return = 2.1166%
- G10 future return = 0.7250%
- Long–short mean return = 1.3916%
- HAC t-statistic = 2.788099
- HAC p-value = 0.005302

The negative IC indicates a reversal-type relationship, while the significant HAC test indicates that the long–short relationship remains statistically significant after accounting for serial dependence.

Among the three Stage 3 combined signals tested so far, Momentum + Trend Strength has the largest observed long–short return and most negative mean IC.

The combination should therefore be carried forward to the multi-factor research stage.

18 Conclusion

The Momentum + Trend Strength combined signal provides strong evidence of a reversal-type relationship with subsequent 20-day stock returns.

The mean Information Coefficient is:

$$IC = -0.068463$$

while the long–short portfolio produces a mean return of:

$$1.3916\%$$

The ordinary long–short t-statistic is:

$$t = 7.793008$$

but the more appropriate HAC / Newey–West test gives:

$$t_{HAC} = 2.788099$$

with:

$$p_{HAC} = 0.005302$$

Therefore, the relationship remains statistically significant at the 5% level after accounting for the overlapping 20-day return structure.

The evidence is consistent with reversal: stocks with lower combined momentum and trend-strength scores tend to have higher subsequent 20-day returns than stocks with higher combined scores.

Among the three Stage 3 combinations tested, this signal shows the largest long–short return and the most negative mean IC.

It is therefore suitable for consideration as a **significant combined reversal signal** in the next stage of the Portfolio Builder research process.

However, an incremental-information test is still required before concluding that trend strength contributes independent predictive information beyond momentum.